F12 4th

Lecture 16

Section 2.5 Zeros of Polynomial Functions

Linear factorization theorem

Rational zero test

Finding rational zeros of a polynomial

Conjugate pairs theorem

Section 2.5 Zeros of Polynomial Functions

Rational functions

Vertical asymptotes

Practice

F124+4

Write the quotient in standard form.

$$\frac{2-5i}{3-4i}$$

solution: $\frac{26}{25} - \frac{7}{25}i$

$$\left(\frac{2-5i}{3-4i}\right)\left(\frac{3+4i}{3+4i}\right) = \frac{6-15i+8i-20i^2}{9+16} = \frac{6-7i-20(-1)}{25}$$

$$= \frac{6-7i+20}{25} = \frac{26-7i}{25} = \frac{26}{25} - \frac{7}{25}i$$

Short-cut

Reminder: $(a+bi)(a-bi) = a^2+b^2$

ex. $(x+3i)(x-3i) = x^2+3^2 = x^2+9$

Lecture 16, Part I: Section 2.5
Zeros of Polynomial Functions

Linear Factorization Theorem

Every polynomial function $f(x)$ of degree $n > 0$ can be factored into n linear factors (not necessarily distinct) of the form

$$f(x) = a_n(x - c_1)(x - c_2)\cdots(x - c_n)$$

where $c_1, c_2, \ldots, c_n$ are complex numbers.

$x^2+4=0$

$x^2=-4$

$x=\pm\sqrt{-4}$

$x=\pm 2i$

That is, every polynomial function of degree $n > 0$ has **exactly n** (not necessarily distinct) **zeros** in the complex number system.

Let $y=0$

4 zeros

ex. Find all zeros of $f(x) = x^4 - 16$ in the complex number system.

$=0$

$f(x) = x^4-16$

$x^4-16=0$

$(x^2+4)(x^2-4)=0$

$(x^2+4)(x+2)(x-2)=0$

$(x+2i)(x-2i)(x+2)(x-2)=0$

$x+2i=0$

$x-2i=0$

$x+2=0$

$x-2=0$

$\boxed{\text{zeros: } -2i, 2i, -2, 2}$

or $x=\pm 2i, \pm 2$

The Rational Zero Test

If the polynomial

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

has integer coefficients, then every rational zero of f has the form $\frac{p}{q}$ (in lowest terms), where p is a factor of a_0 and q is a factor of a_n.

Finding the Rational Zeros of a Polynomial

1. LIST POSSIBLE ZEROS. List all possible rational zeros using the Rational Zero Test.

If easy numbers, use factor theorem $f(x)=0$

2. DIVIDE. Use synthetic division to evaluate the polynomial at each of the candidates for rational zeros that you found in Step 1. When the remainder is 0, note the quotient you have obtained.

3. REPEAT. Repeat Steps 1 and 2 for the quotient. Stop when you reach a quotient that is quadratic or factors easily, and use the quadratic formula or factor to find the remaining zeros.

① Possible rational zeros: $\dfrac{\text{factors of } 6}{\text{factors of } 1} = \dfrac{\pm 1, \pm 2, \pm 3, \pm 6}{\pm 1}$

thus $\pm 1, \pm 2, \pm 3, \pm 6$

<u>ex.</u> Find the rational zeros of the polynomial:

1) $f(x) = 1x^3 - 4x^2 + x + 6 = (x-1)(\quad)(\quad)$

Can use synthetic div.

$$\begin{array}{r|rrrr} 1 & 1 & -4 & 1 & 6 \\ \hline & & & & \end{array}$$

or

Factor theorem $f(1) = 1 - 4 + 1 + 6 \neq 0$

$f(-1) = -1 - 4 - 1 + 6 = 0$

thus -1 is a zero and $(x-(-1))$ is a factor.

Now use syn. div to factor

$$\begin{array}{r|rrrr} -1 & 1 & -4 & 1 & 6 \\ & & -1 & 5 & -6 \\ \hline & 1 & -5 & 6 & 0 \\ & x^2 & x & & \end{array}$$

$f(x) = x^3 - 4x^2 + x + 6 = (x-(-1))(x^2 - 5x + 6) =$

$f(x) = (x+1)(x-2)(x-3)$

zeros: $\boxed{x = -1,\ x = 2,\ x = 3}$

possible rational zeros: $\frac{\pm 1, \pm 2, \pm 4}{\pm 1}$

thus $\pm 1, \pm 2, \pm 4$

2) $g(x) = 1x^4 - 2x^3 - 3x^2 + 8x - 4$

Try $x = 1$ $\quad g(1) = 1 - 2 - 3 + 8 - 4 = 0$ ✓ thus $x - 1$ is a factor

$$\begin{array}{c|ccccc} & 1 & -2 & -3 & 8 & -4 \\ 1 & & 1 & -1 & -4 & 4 \\ \hline & 1 & -1 & -4 & 4 & 0 \\ & x^3 & x^2 & x & c & r \end{array}$$

can repeat the process

$f(x) = (x-1)(x^3 - x^2 - 4x + 4)$

$f(x) = (x-1)[(x^3 - x^2) - (4x - 4)]$

$f(x) = (x-1)[x^2(x-1) - 4(x-1)]$

$f(x) = (x-1)[(x-1)(x^2-4)]$

$f(x) = (x-1)^2(x+2)(x-2)$

$\boxed{\text{zeros: } x = 1, \quad x = -2, \quad x = 2}$

multiplicity: two, one, one

even, odd, odd

Recall [redacted] $(a-bi)(a+bi) = a^2+b^2$

Conjugate Pairs Theorem

If $a + bi$ $(b \neq 0)$ is a zero of a polynomial function f with real coefficients, then the complex conjugate $a - bi$ is also a zero of f. $a+bi,\ a-bi$

Corollary: A polynomial f of odd degree with real coefficients has at least one real zero.

<u>ex.</u> Find a polynomial of degree 4 with real coefficients that has 0, -1 and $1 + i$ as zeros. also $1-i$

$$f(x) = (x-0)(x-(-1))(x-(1+i))(x-(1-i))$$

$$f(x) = x(x+1)(x-1-i)(x-1+i)$$

can use distributive property to multiply it out. wicked short-cut

$$f(x) = x(x+1)[((x-1)-i)((x-1)+i)]$$

$$f(x) = x(x+1)[(x-1)^2 + 1^2] =$$

$$f(x) = x(x+1)[x^2-2x+1+1] = x(x+1)(x^2-2x+2) =$$

$$f(x) = (x^2+x)(x^2-2x+2) = x^4-2x^3+2x^2 + x^3-2x^2+2x$$

$$f(x) = x^4 - x^3 + 2x \qquad x^4 - x^3 + 2x$$

Note: Many answers are possible.
$f(x) = 2(x^4-x^3+2x)$

ex. Find all zeros of $f(x) = 2x^4+5x^3+5x^2+20x-12$ given that $-2i$ is a zero of f. thus $2i$ is also a zero.

Factors: $(x-(-2i))\ (x-2i) = (x+2i)(x-2i)$

$= x^2+4$ is a factor

$$
\begin{array}{r}
2x^2+5x-3 \\
x^2+0x+4 \overline{\big) 2x^4+5x^3+5x^2+20x-12} \\
-(2x^4 \quad 0x^3 \quad +8x^2) \\
\hline
5x^3-3x^2+20x-12 \\
-(5x^3+0x^2+20x) \\
\hline
-3x^2+0x-12 \\
-(-3x^2+0x-12) \\
\hline
0
\end{array}
$$

thus $f(x) = (x^2+4)(2x^2+5x-3)$

$f(x) = (x^2+4)(2x-1)(x+3)$

$\boxed{\text{zeros: } \pm 2i, \tfrac{1}{2}, -3}$

Lecture 16, Part II: Section 2.6
Rational Functions

<u>Def.</u> A **rational function** is a function of the form

$$f(x) = \frac{p(x)}{q(x)}$$

where p and q are polynomial functions and q is not the zero polynomial.

Domain of a rational function: is all real numbers, except where $q(x)=0$

The **zeros** of a rational function: are where

$$\frac{p(x)}{q(x)} = 0.$$

let $q=0$ to find the x-intercepts.

zeros.

Domain: Factor the denominator to find the zeros of the denominator.

All reals except

D: $x \neq 0, 1$

ex. Find the domain and zeros.

1) $f(x) = \dfrac{x^2}{x^2 - x} = \dfrac{x^2}{x(x-1)}$

Find the zeros (x-intercepts), so let $y=0$.

$\dfrac{x^2}{x^2-x} = 0 \rightarrow x^2=0$, thus zero is $x=0$

however $x=0$, is not in the domain, so no zeros.

2) $f(x) = \dfrac{3x}{x^2+1}$

D: x^2+1 doesn't factor over the reals.

$x^2+1=0$

$x^2=-1$ no real solution,

thus domain is $\boxed{\text{all reals.}}$

Find the zeros

$3x=0$

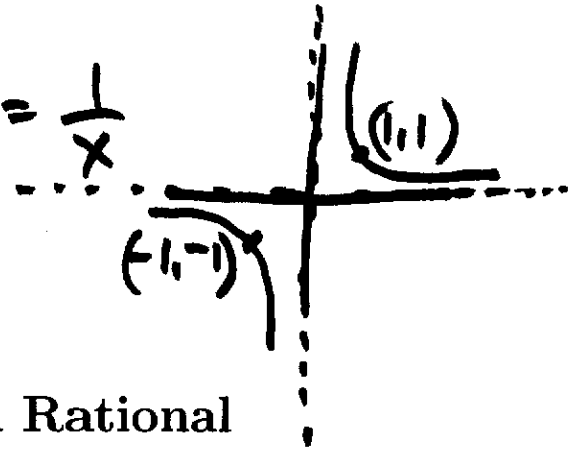
$\boxed{x=0}$

3) $f(x) = \dfrac{x^2+4x+3}{x^2-9} = \dfrac{(x+3)(x+1)}{(x+3)(x-3)}$

D: All reals, except $x \neq \pm 3$

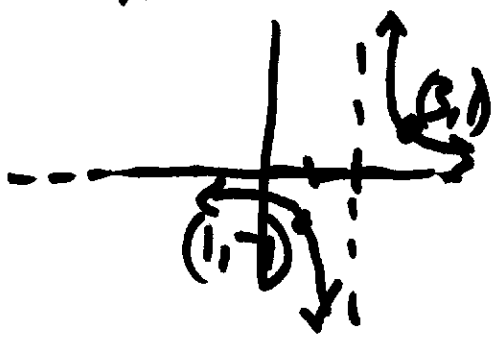
zeros: Numerator $\cancel{x=-3}$, $x=-1$

① $f(x) = \frac{1}{x}$



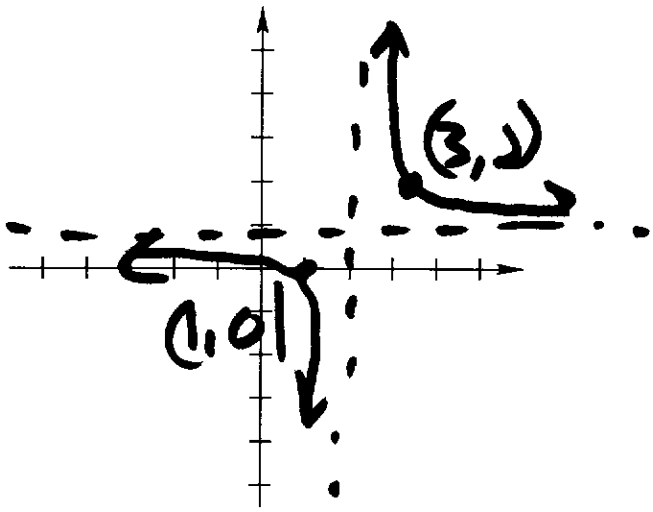
Using Translation to Graph a Rational Function

② $f(x) = \frac{1}{x-2} \rightarrow 2$



<u>ex.</u> Graph $f(x) = \dfrac{1}{x-2} + 1$

③



Vertical Asymptotes

<u>**Def.**</u> The line $x = a$ is a **vertical asymptote (VA)** of the graph of rational function $f(x)$ if

$|f(x)| \rightarrow \infty$ when x approaches a

from the left or from the right.

4th
to be finished on W

NOTE:

1. The line $x = a$ is a VA of the graph of $f(x)$ if a is the zero of the **simplified** denominator.

2. A graph DOES NOT cross its vertical asymptote(s).

ex. Find each vertical asymptote of the function.

1) $f(x) = \dfrac{x+2}{x^2-9}$

2) $g(x) = \dfrac{2}{x^2+4}$

3) $h(x) = \dfrac{x^2-2x}{x^2-3x+2}$